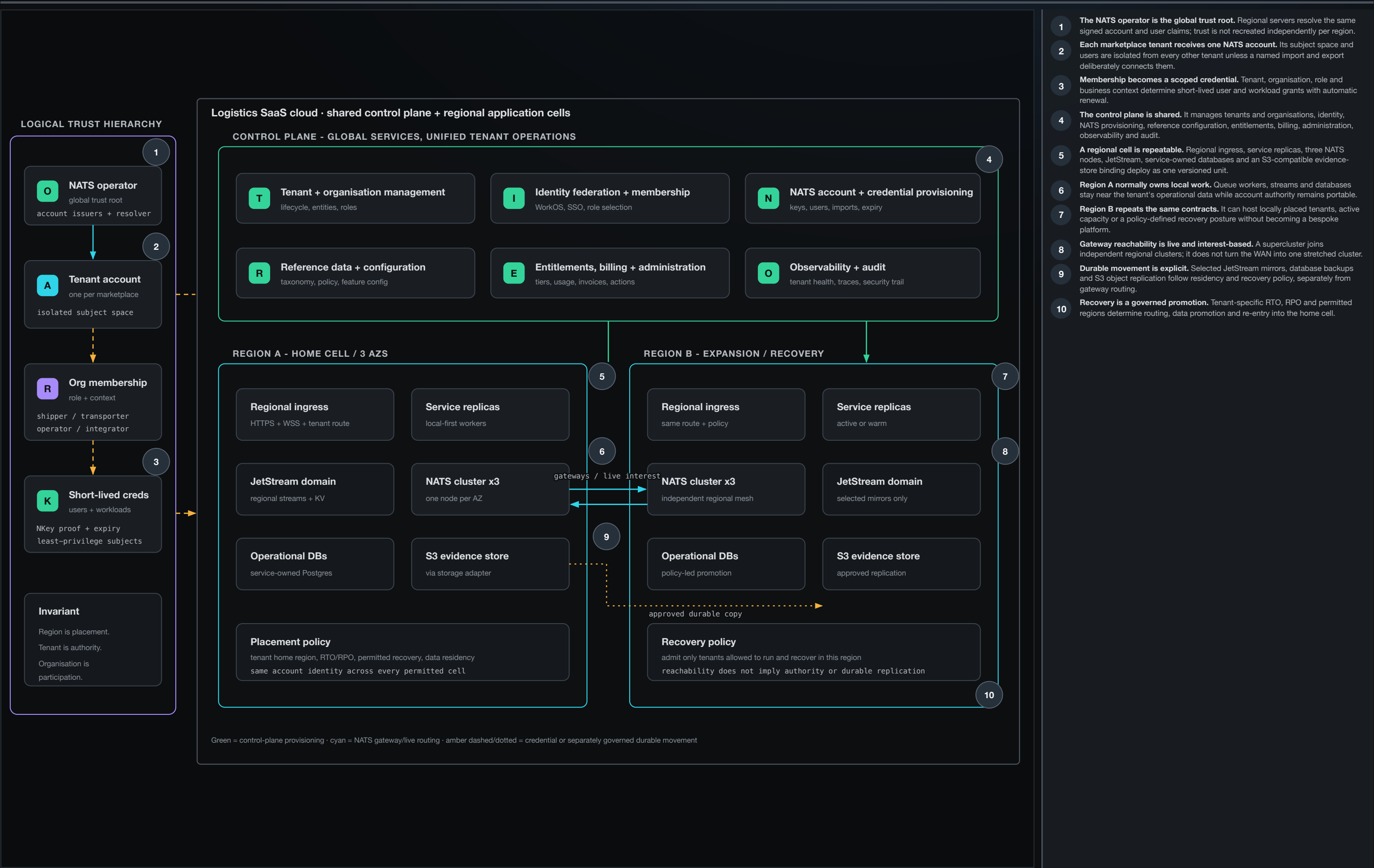


Marketplace Tender-to-Settlement on NATS

This architecture shows how shippers, transporters, drivers and external systems collaborate through a governed logistics marketplace. Numbered flows follow the AWS reference-architecture convention; the services and guarantees remain the proposed NATS-based platform model.

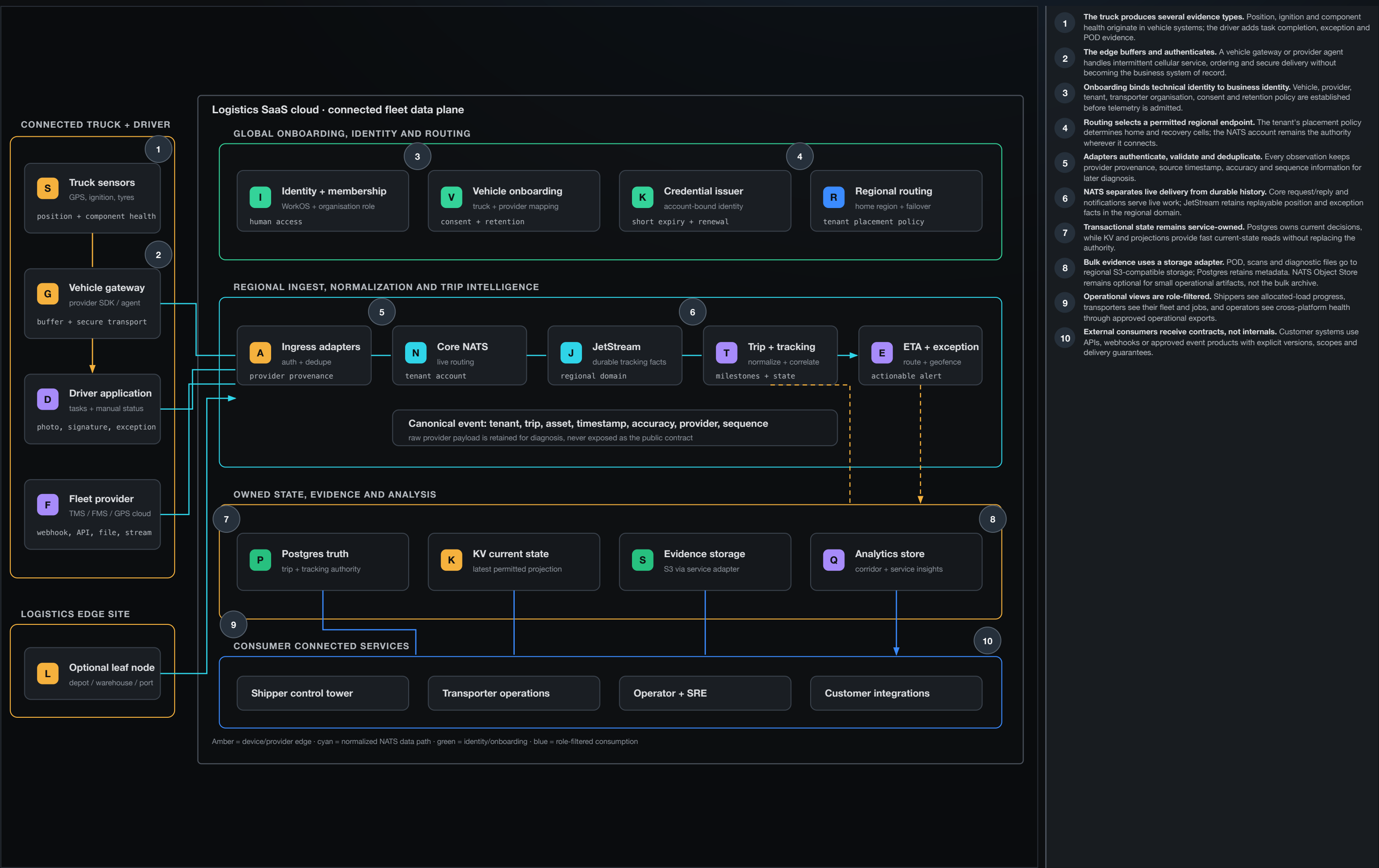
Multi-region Tenant Authority and Deployment

This architecture separates the logical trust hierarchy and shared control plane from two repeatable physical regional cells. A tenant account can be resolved in permitted regions without making region part of tenant identity.



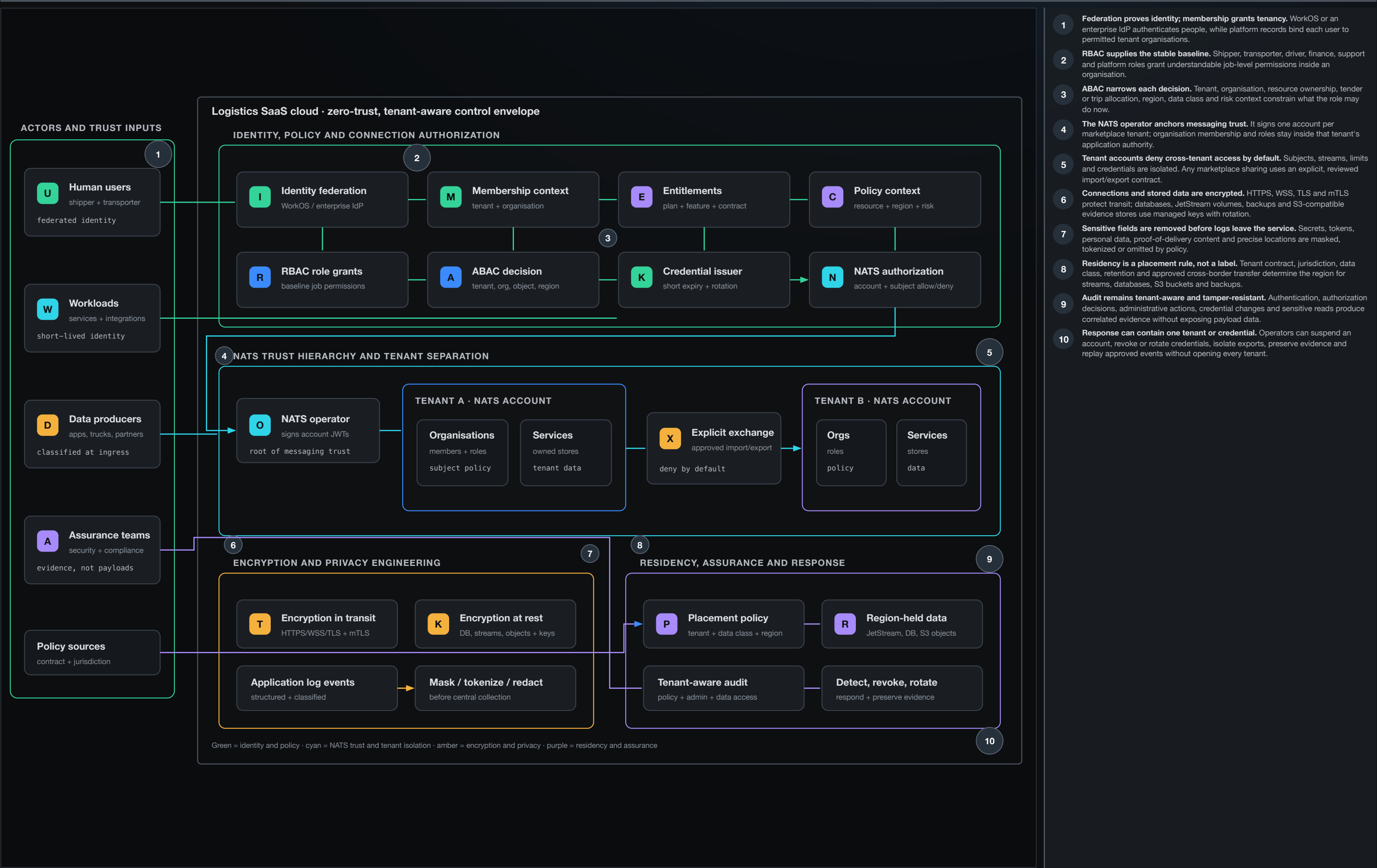
Connected Fleet Data Gathering, Processing and Use

This architecture follows vehicle data from truck sensors, driver applications and fleet providers through secure onboarding, regional NATS ingestion, normalization, state projection and role-filtered consumer services.



Security, Privacy and Regulatory Controls

This control view places identity, authorization, tenant separation, encryption, privacy, residency and audit around the same NATS-based logistics platform. It defines architectural requirements and evidence points; it does not claim certification or replace jurisdiction-specific legal and security review.



GLOBAL LOGICAL CONTROL PLANE - AUTHORITY AND PLACEMENT METADATA, NO REGULATED FREIGHT PAYLOAD

1 Identity + membership
WorkOS + organisation roles
tenant context

2 Tenant lifecycle
authority, tier, countries
activate / suspend

3 NATS operator
account JWTs + resolver
global trust root

4 Placement directory
home + permitted recovery
tenant -> regional cell

5 Provision + observe
services, policy, SRE
desired vs observed

place + provision shared-region tenants

3 SHARED CAPE TOWN CELL - AWS AF-SOUTH-1 / 3 AZS

Regional ingress
HTTPS + WSS
tenant routed

Service replicas
local workers
pool or silo

NATS cluster x3
one server / AZ
regional routes

JetStream AFRICA
R=3 streams + KV
Cape Town only

Service Postgres
tenant scoped
regional backup

Evidence adapter
S3-compatible
regional objects

5

7

South Africa + eligible regional tenants
tenant authority stays distinct from country
home=af-south-1 · shared or isolated tier

4 BOTSWANA SOVEREIGN COUNTRY CELL - REGULATED TENANTS

Local identity + tenant control + ingress
Botswana credential, policy and endpoint authority

NATS cluster x3
in-country nodes
local client routes

Service replicas
Botswana workers
tenant silo

JetStream BW
R=3 streams + KV
no external mirrors

Service Postgres
local authority
local replicas

Evidence objects
S3-compatible
local adapter + keys

Logs + audit
masked in-country
local SIEM target

8

Local KMS/HSM · backups · recovery site

6 SHARED SYDNEY CELL - AWS AP-SOUTHEAST-2 / 3 AZS

Regional ingress
HTTPS + WSS
tenant routed

Service replicas
local workers
pool or silo

NATS cluster x3
one server / AZ
regional routes

JetStream AU
R=3 streams + KV
Sydney only

Service Postgres
tenant scoped
regional backup

Evidence adapter
S3-compatible
regional objects

6

9 COUNTRY-RESIDENCY POLICY GUARDRAILS - BOTSWANA EXAMPLE

Placement: home=BW · recovery=BW only
compute, messages and all durable stores

Cross-border default: DENY
no gateway, mirror, backup, object or log export

Exception: explicit regulatory approval
masked / aggregate operational metadata only

Cape Town ↔ Sydney NATS gateways; Botswana subjects excluded

- ## CONNECTED FREIGHT REFERENCE ARCHITECTURE